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Thesis Presentation

NETWORK INFERENCE FOR HOSPITAL INFECTION TRANSMISSION

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Agenda Overview

01 Introduction

02 Literature Review

03 Methodology

04 Simulation

05 Result

06 Conclusion / Discussion



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Health Background

Carbapenemase-producing Enterobacteriaceae (CPE) is a major health threat that has been increasingly since 2019:

- High morbidity and mortality
- High healthcare costs
- Patients have to be managed under contact precautions

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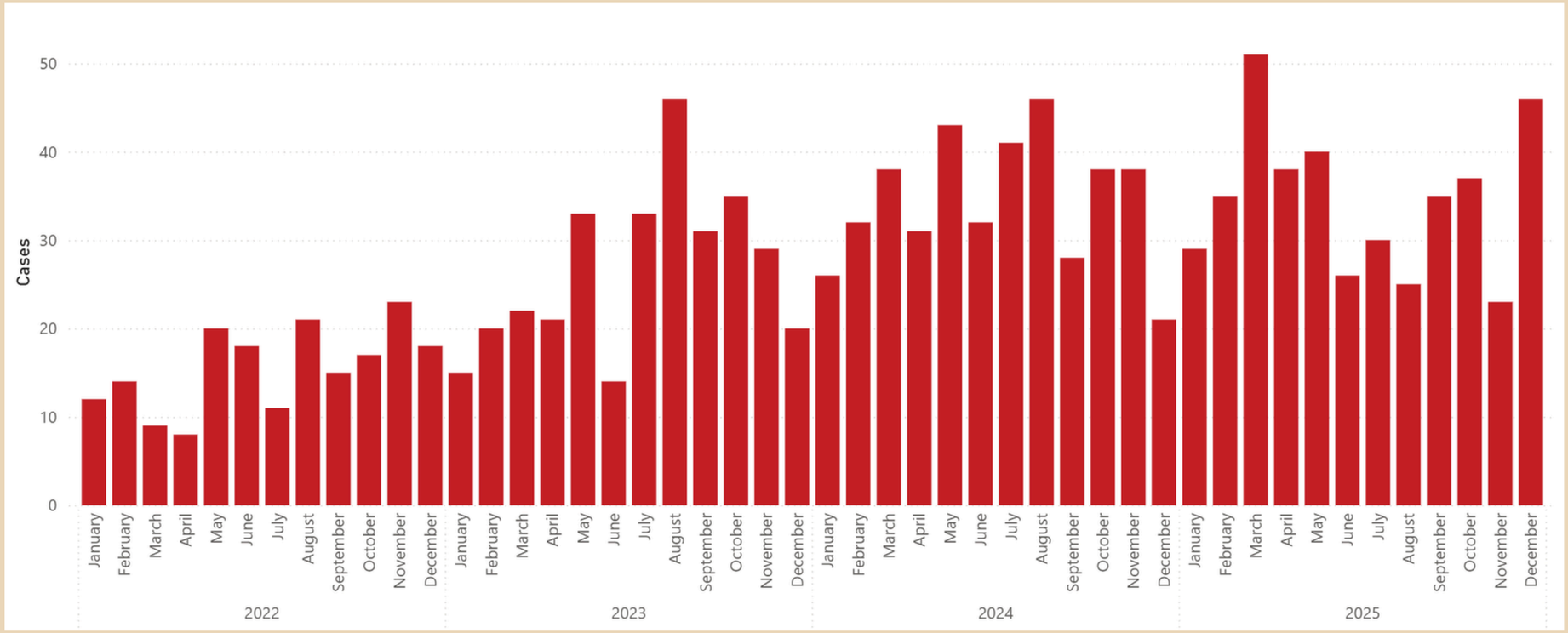
Motivation

CPE infections in Victoria have continued to rise (since 2019) - ICU

The current research about CPE

- Focuses on patient-level risk factors
- No transmission pathways

→ Hard to identify high-risk wards or units where transmission is likely to occur



Department of Health, Victoria (2026)

Motivation

Investigate the pathways of inter-ward transmission of CPE infections by analysing ward-level CPE count data across 3 different campuses from a single healthcare provider in Victoria, Australia

Infer the conditional dependence structure between wards

Apply Gaussian graphical model (GGMs)



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Mathematical Background



Gaussian graphical models provide a flexible framework for representing complex dependence structures among multivariate random variables.

Nodes and Edges

- Variables are nodes
- Edges encode conditional (in)dependence relationships

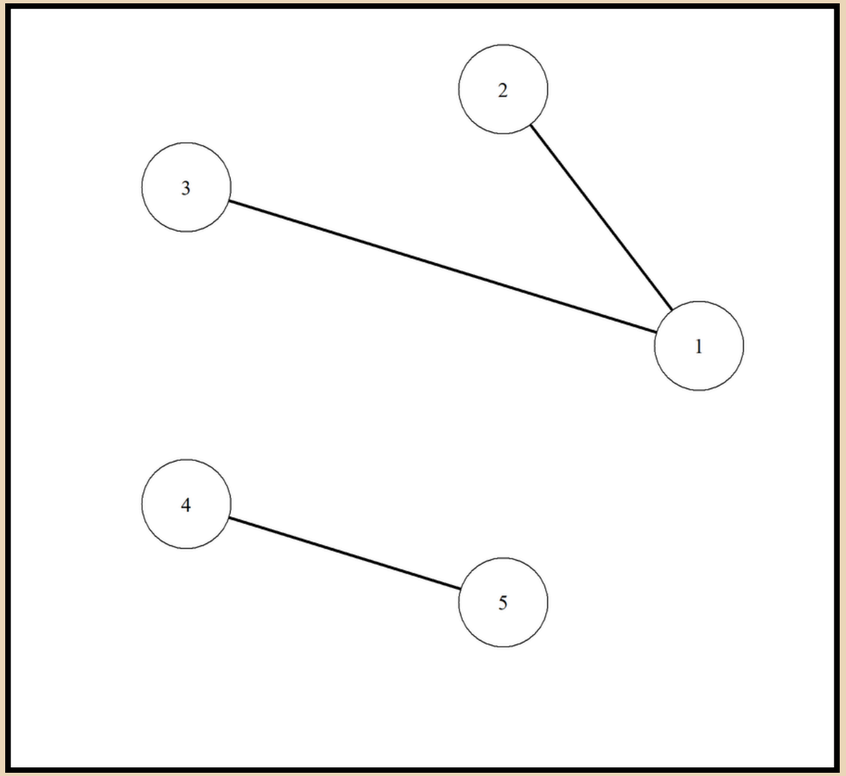
Assumptions

- Multivariate Normality
- No temporal autocorrelation

Spike-and-slab

- Define the sparsity in Bayesian graph (Wang 2015)
- Improve the scalability in high-dimensional settings (Jewson et al. 2024)

Gaussian Graphical Model (GGM)



Nodes and Edges

- The nodes are 1, 2, 3, 4,5
- Edges represents for conditional (in)dependence relationships
- 1 – 2 ; 1 – 3 ; 4 – 5

Interpretation

- 1 is conditionally dependent with 2 and 3.
- 4 is conditionally dependent with wards 5
- There is no direct conditional dependence between (2,3) ; (3,4) ; (3,5) ; (1,4) ; (1;5) ; ...

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Notation

Observation

Obersved count

$Y_{t,j}$: The number of cases in ward j, time t

Time: $t = 1, \ldots, T$

Ward / Variables: $j = 1, \ldots, P$

Conditional Dependence Structure

Graph:

$G = (V, E)$:set of edges where

Nodes:

$V, E = \{1, \ldots, P\}$: set of nodes

The goal of this study is to estimate the graph from the observed count data while accounting for zero-inflation and temporal dynamics

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Literature Review

Zero-Inflated Poisson Log-Normal (ZIPLN)

Batardière et al. (2025)

Observed count

$$Y_{t,j} = \begin{cases} 0, \text{ if } W_{t,j} = 0, \\ Poisson(exp(H_{t,j})) \text{ if } W_{t,j} = 1. \end{cases}$$

Latent Bernoulli

$$W_{t,j} \sim Bernoulli(\pi_{t,j})$$

Latent Gaussian vector

$$H_t = (H_{t,1}, \ldots, H_{t,P})^T \sim N(\mu_t, \Sigma_H)$$

Interpretation

Ω_H^{-1} : Capture the conditional dependence graph



Assume temporal independence across time points

Extension

Introducing autoregressive latent:

- Poisson intensity
- Zero-inflation

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Methodology

Zero-Inflated Poisson

$Y_{t,j} \sim \mathrm{ZIP}(\lambda_{t,j}, \pi_{t,j})$

$\lambda_{t,j} > 0$: Poisson intensity

$\pi_{t,j} \in (0,1)$: Zero-inflation structure

Probability mass function

$$P(Y_{t,j} = y_{t,j}) = \begin{cases} \pi_{t,j} + (1-\pi_{t,j})e^{-\lambda_{t,j}}, & \text{if } y_{t,j} = 0, \\ (1-\pi_{t,j})\frac{e^{-\lambda_{t,j}}\lambda_{t,j}^{y_{t,j}}}{y_{t,j}!}, & \text{if } y_{t,j} > 0. \end{cases}$$

Dynamic ZIPLN

Independence:

$$\log(\lambda_{t,j}) = \beta_{0,j} + S_{t,j},$$

$$logit(\pi_{t,j}) = Z_{t,j},$$

$\beta_{0,j}$: Baseline log-intensity

$S_{t,j}$: Latent intensity

$Z_{t,j}$: Latent zero-inflation

→ Extend the latent process to capture the temporal autocorrelation

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Latent Process

Autoregressive process of order 1

Poisson-Intensity Latent

$$S_{t,j} = (1-\phi_j)c_{sj} + \phi_j S_{t-1,j} + \varepsilon^{(s)}_{t,j}, \qquad \varepsilon^{(s)}_{t,j} \sim N(0, \sigma^2_{sj}).$$

$$S_{t,j} \mid S_{t-1,j} \sim \text{Normal}\left((1-\phi_j)c_{sj} + \phi_j S_{t-1,j}, \sigma^2_{sj}\right), \quad \text{for } t = 2, \ldots, T$$

c_{sj} : the stationary mean
$\phi_j \in (-1, 1)$: ward specific AR(1)
σ^2_{sj} : innovation variance

Zero-Inflation Latent

$$Z_{t,j} = (1-\rho_j)c_{zj} + \rho_j Z_{t-1,j} + \varepsilon^{(z)}_{t,j}, \qquad \varepsilon^{(z)}_{t,j} \sim N(0, \sigma^2_{zj}).$$

$$Z_{t,j} \mid Z_{t-1,j} \sim \text{Normal}\left((1-\rho_j)c_{zj} + \rho_j Z_{t-1,j}, \sigma^2_{zj}\right), \quad \text{for } t = 2, \ldots, T$$

c_{zj} : the stationary mean
$\rho_j \in (-1, 1)$: ward specific AR(1)
σ^2_{sj} : innovation variance

→ Capture stochastic temporal autocorrelation, not only a strict linear trend

Post-Hoc Gaussian Graphical Model (GGM) Network Inference

Latent Innovations

Isolate true contemporaneous dependencies by extracting the latent innovations:

$$r_{t,j} = S_{t,j} - \mathbb{E}[S_{t,j} \mid S_{t-1,j}] = S_{t,j} - ((1-\phi_j)c_{s,j} + \phi_j S_{t-1,j})$$

$$\hookrightarrow \mathbf{r}_t = (r_{t,1}, \ldots, r_{t,P})$$

Precision Matrix

Assume: $\hat{\mathbf{r}}_t \sim \mathcal{N}_P(\mathbf{0},\ \boldsymbol{\Omega}^{-1}), \quad t = 2, \ldots, T$

→ $\Omega_{j,k} = 0 \iff \hat{r}_{t,j} \perp \hat{r}_{t,k} \mid \hat{\mathbf{r}}_{t,\setminus\{j,k\}}$

→ Apply GGM (Sulem et al. (2025))

→ $G = (V, E)$

→ Applying GGM to determine whether an association between two wards remains after accounting for the counts observed in all other wards (rather than simple pairwise correlations)

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Simulation

Data-Generating Process

Latent process:

- $S_t = (S_{t,1}, \ldots, S_{t,P})^\top$
- $Z_t = (Z_{t,1}, \ldots, Z_{t,P})^\top$
- Each is generated independently

Graph structure:

- $\epsilon_t = (\epsilon_{t,1}, \ldots, \epsilon_{t,P})^\top \sim N_P(0, \Sigma)$
- $\Omega = \Sigma^{-1}$
- Off-diagonal elements encode the edge

Scenario 1
(P = 10, T = 500)
No auto-correlation

$\Omega_{1,5} = \Omega_{5,1} = -0.8$

$\Omega_{2,3} = \Omega_{3,2} = 0.8$

$\rho = 0$

$\phi = 0$

$c_{Z,j} = 0$

$c_{S,j} = 0$

Scenario 2
(P = 10, T = 500)
Temporal dependence

$\Omega_{1,5} = \Omega_{5,1} = -0.8$

$\Omega_{2,3} = \Omega_{3,2} = 0.8$

$\rho = 0.8$

$\phi = 0.8$

$c_{Z,j} = 0.5$

$c_{S,j} = 1$

Scenario 3
(P = 30, T = 100)
Temporal dependence

$10\ \Omega_{j,k}$

$\rho = 0.8$

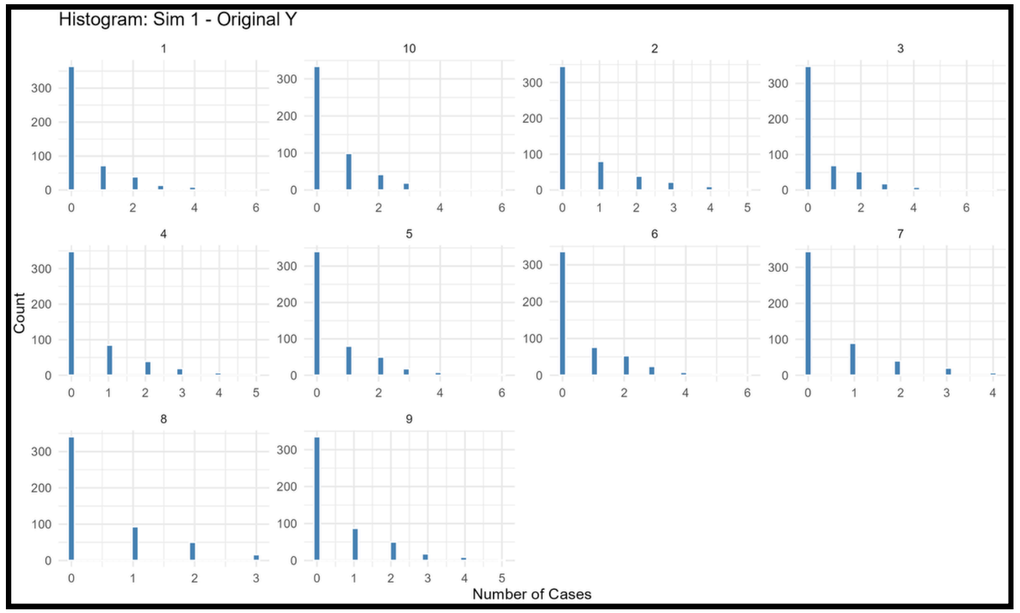
$\phi = 0.8$

$c_{Z,j} = 0.5$

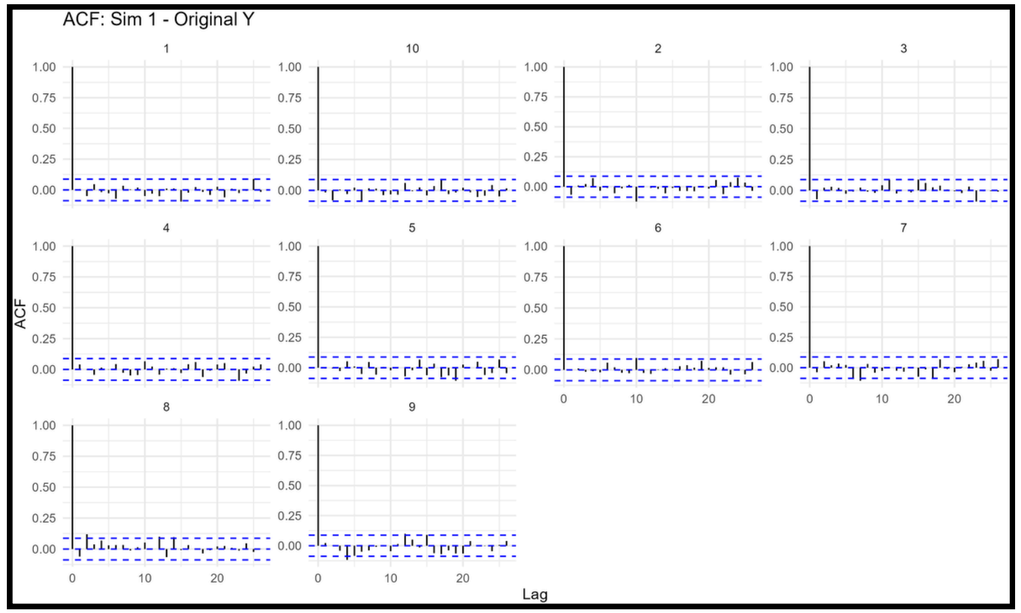
$c_{S,j} = 1$

Simulation

Scenario 1: (P = 10, T = 500)
No auto-correlation



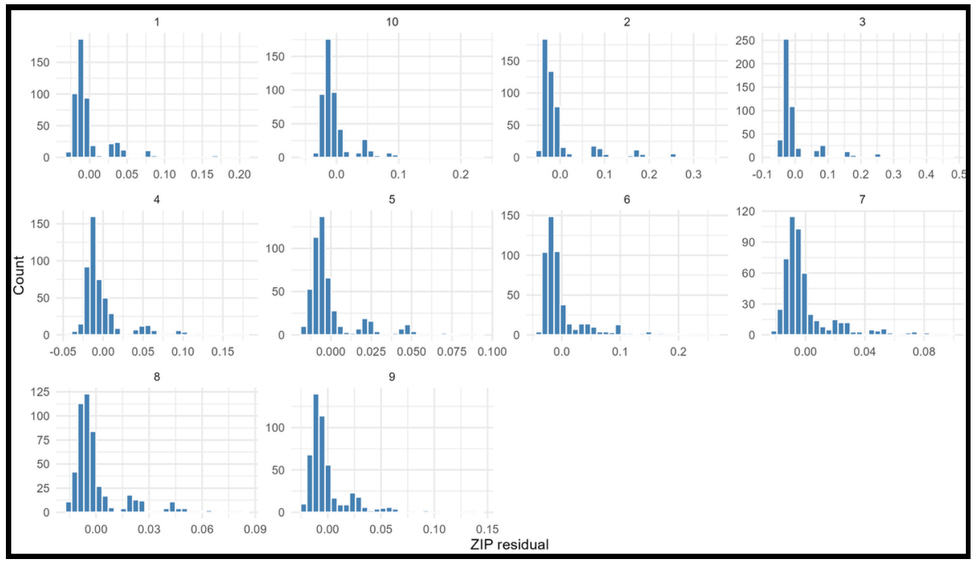
Raw observed count: Histogram



Raw observed count: ACF plot

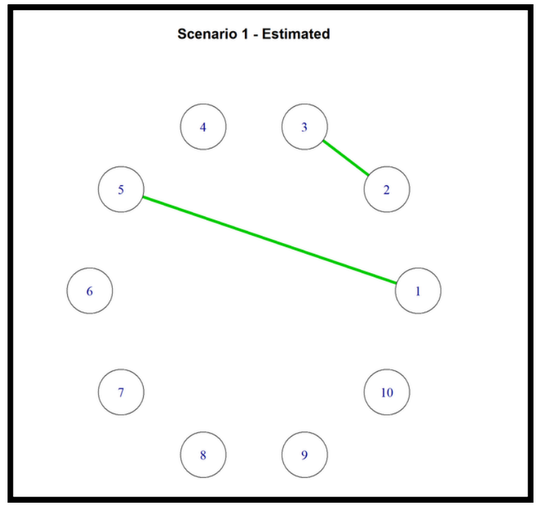
Simulation

Scenario 1: (P = 10, T = 500)
No auto-correlation

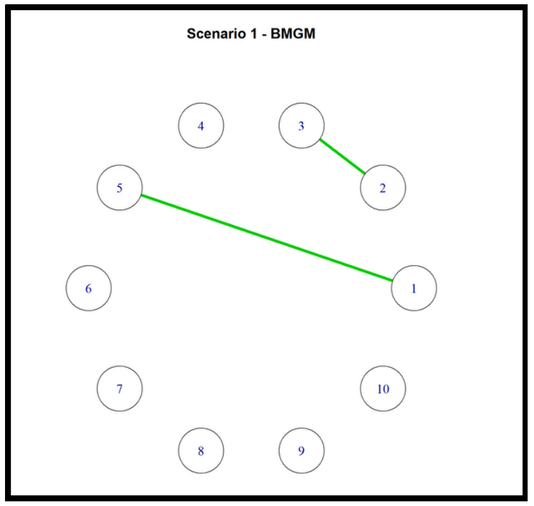


Innovation latent: Histogram

GGM



Estimated graph



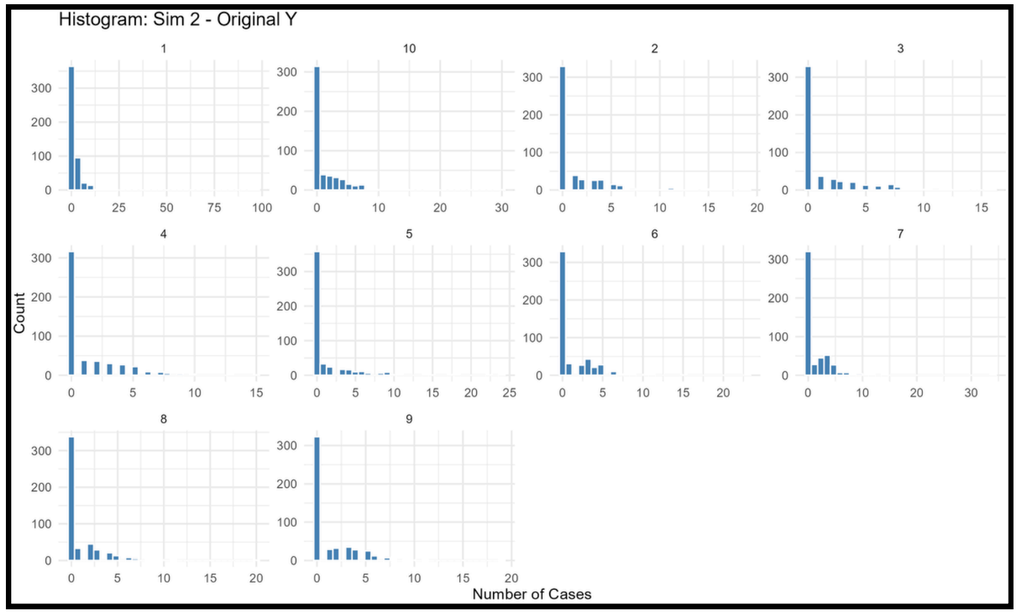
BMGM graph

(Florez et al. (2025))

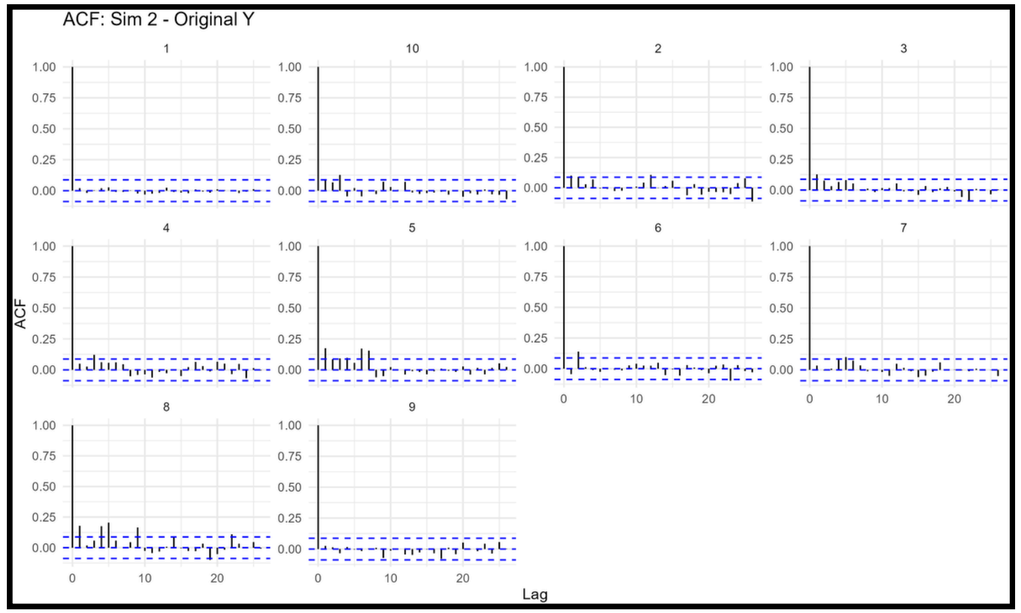
- Accounts for non-Gaussian but not include time-series
- Uses a composite likelihood

Simulation

Scenario 2: (P = 10, T = 500)
Temporal dependence



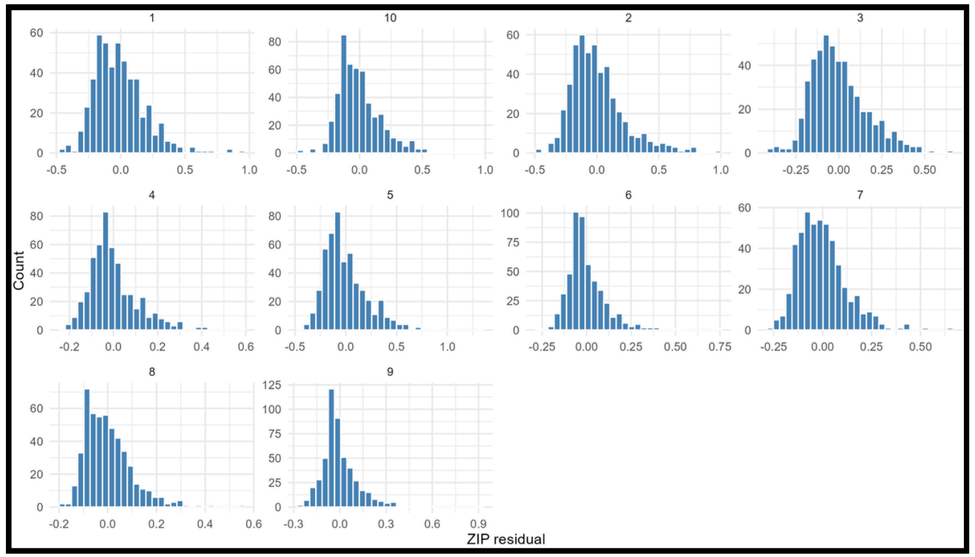
Raw observed count: Histogram



Raw observed count: ACF plot

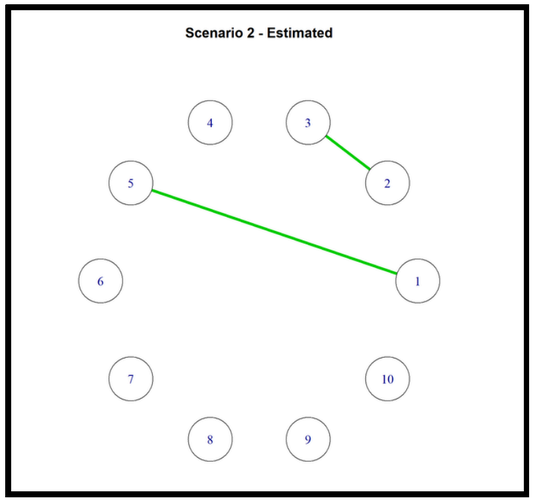
Simulation

Scenario 2: (P = 10, T = 500)
Temporal dependence

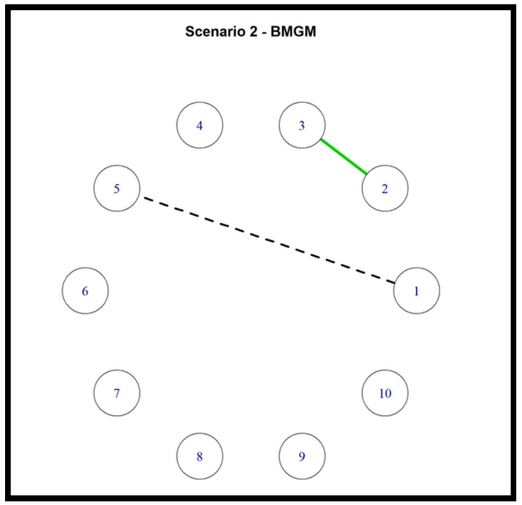


Innovation latent: Histogram

GGM



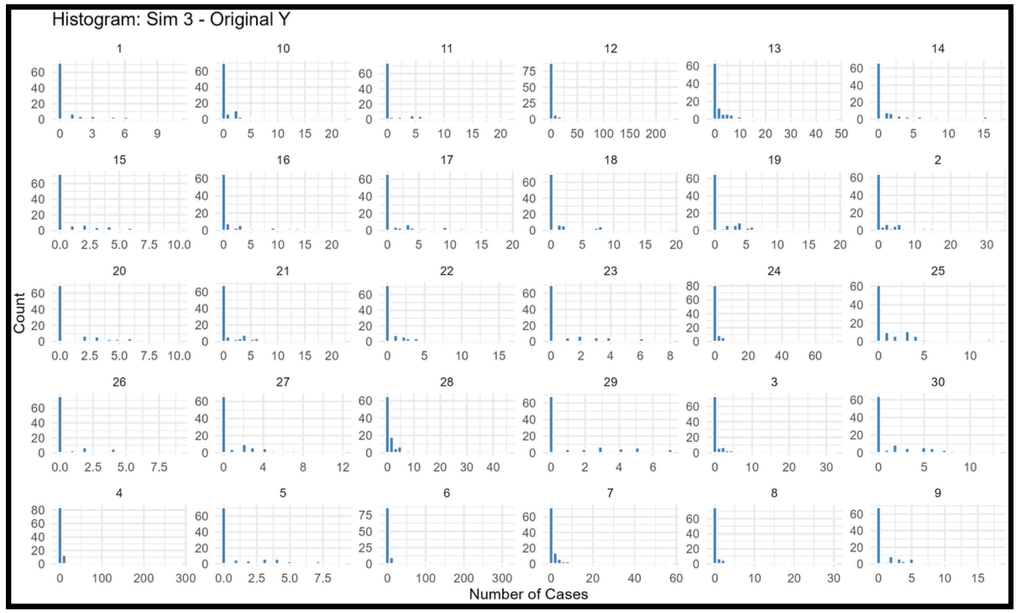
Estimated graph



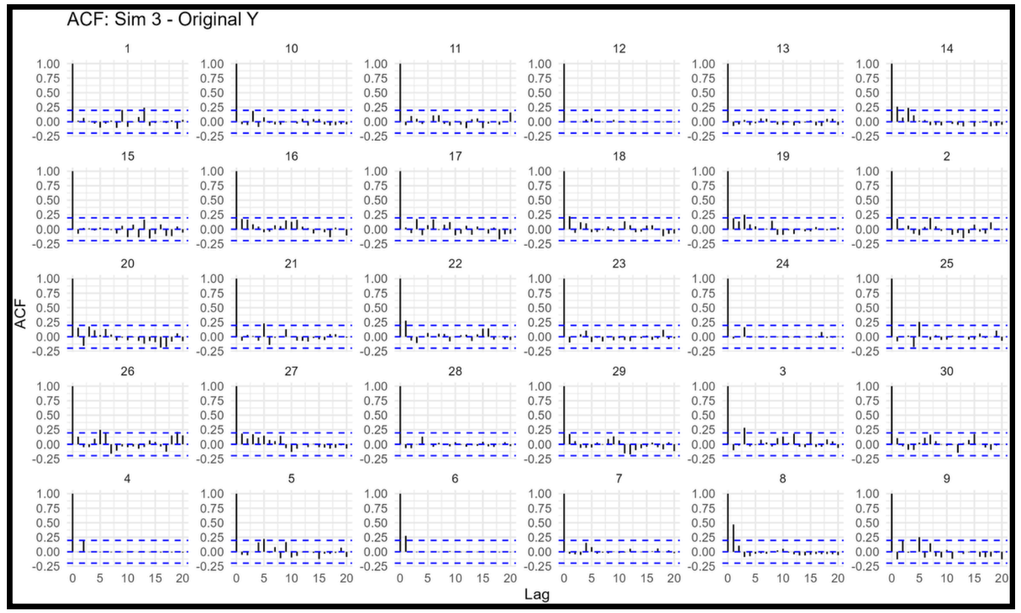
BMGM graph
(Florez et al. (2025))

Simulation

Scenario 3: (P =30, T = 100)
Temporal dependence



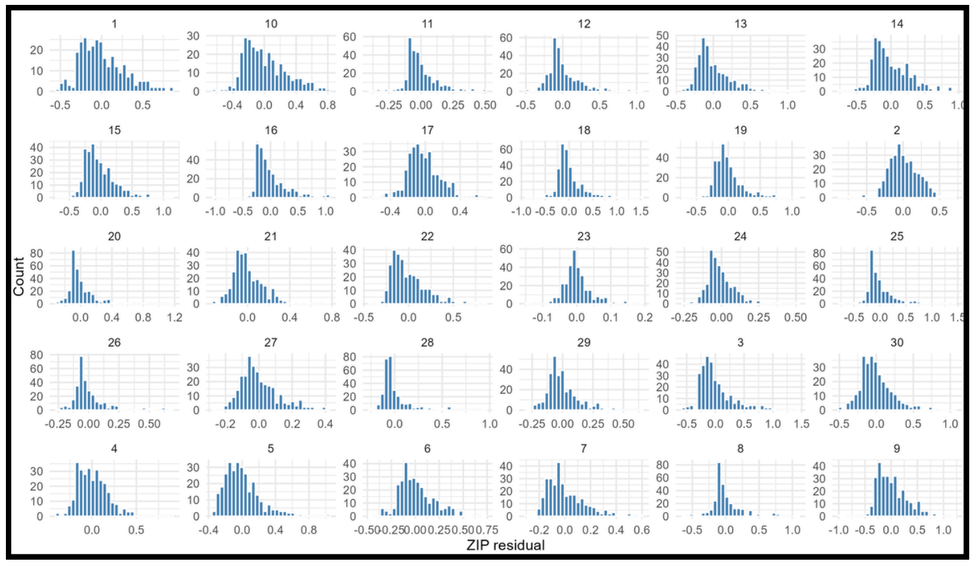
Raw observed count: Histogram



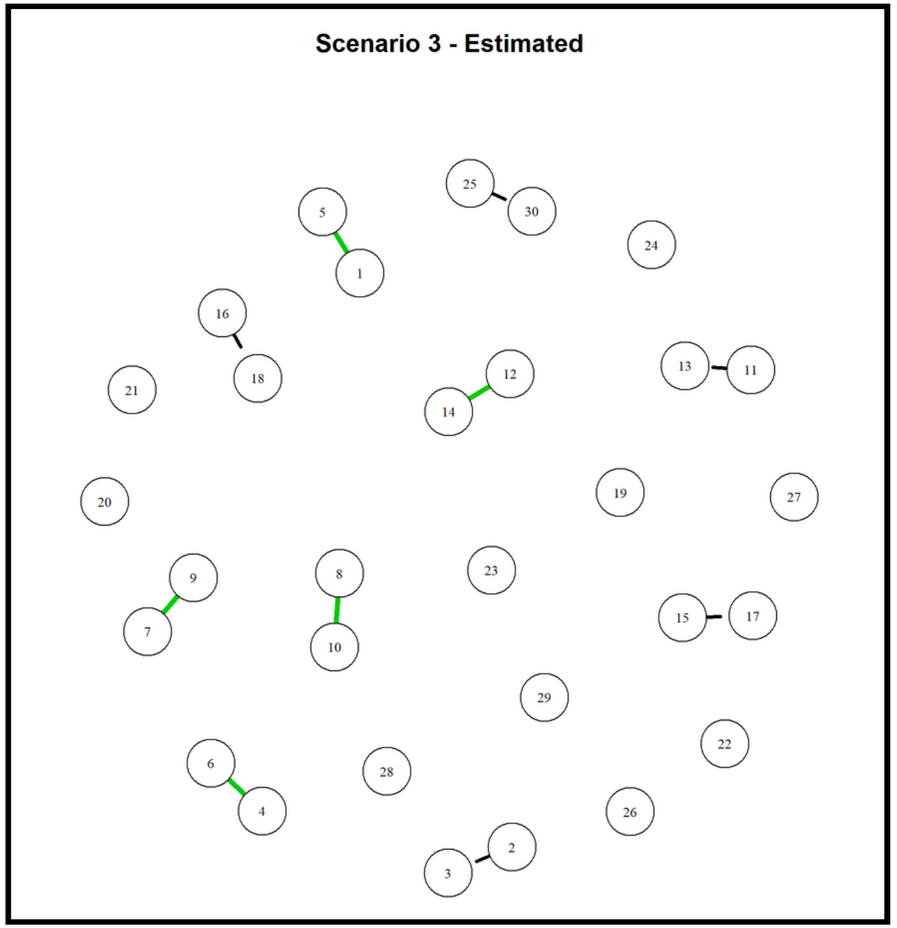
Raw observed count: ACF plot

Simulation

Scenario 3: (P =30, T = 100)
Temporal dependence



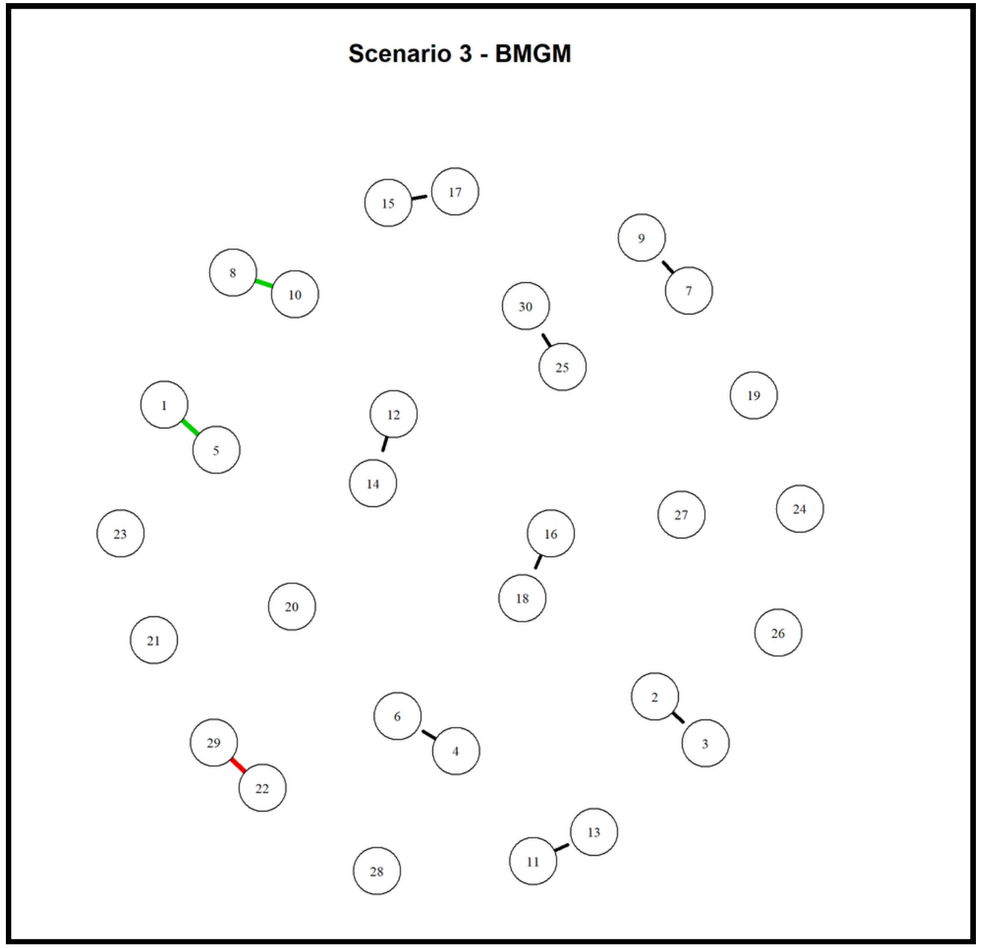
GGM



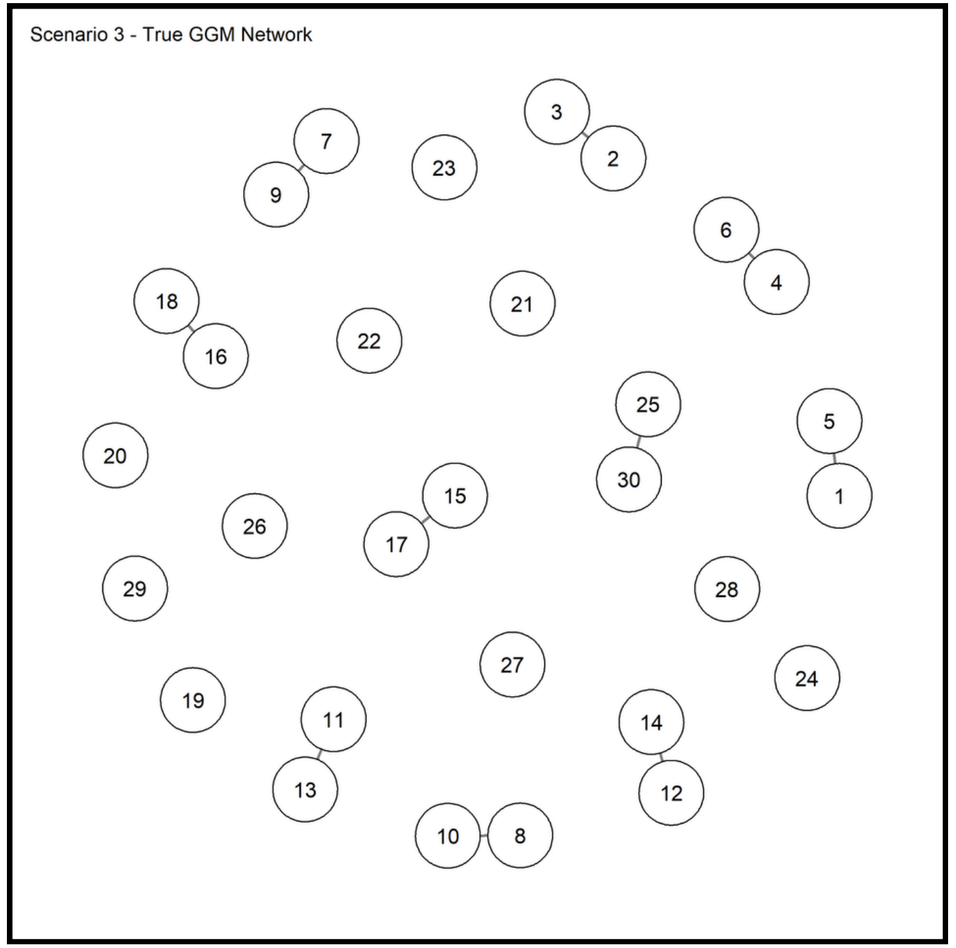
Innovation latent: Histogram

Estimated graph (5/10 correct)

Simulation



BMGM graph
(2 correct, 1 false)
(Florez et al. (2025))



True graph
(10 edges)

Simulation

Parameter Recovery Performance Across Simulation Scenarios

Scenario	Parameter	True Value	Posterior Mean	Bias	RMSE	95% CI
Scenario 1	ρ	0.0	0.032	0.032	0.497	[-0.784, 0.905]
	ϕ	0.0	0.120	0.120	0.504	[-0.741, 0.871]
Scenario 2	ρ	0.8	0.417	-0.383	0.702	[-0.673, 0.996]
	ϕ	0.8	0.775	-0.025	0.119	[0.578, 0.900]
Scenario 3	ρ	0.8	0.516	-0.284	0.623	[-0.567, 0.997]
	ϕ	0.8	0.739	-0.061	0.268	[0.429, 0.924]

Generated by posterior predictive check in stan

↓

95% CI cover the true values

→ Be able to capture the stochastic temporal trend

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Application: Carbapenemase-producing Enterobacteriaceae (CPE) in Victoria

Data is collected from **Alfred Health**, including 3 hospitals:

- Wards were encoded
- This study was approved by Alfred Health Human Research Ethics Board (Local reference number 445/21)

Variables:

- Encoded Wards | Time (month-year) | Number of CPE cases in this month

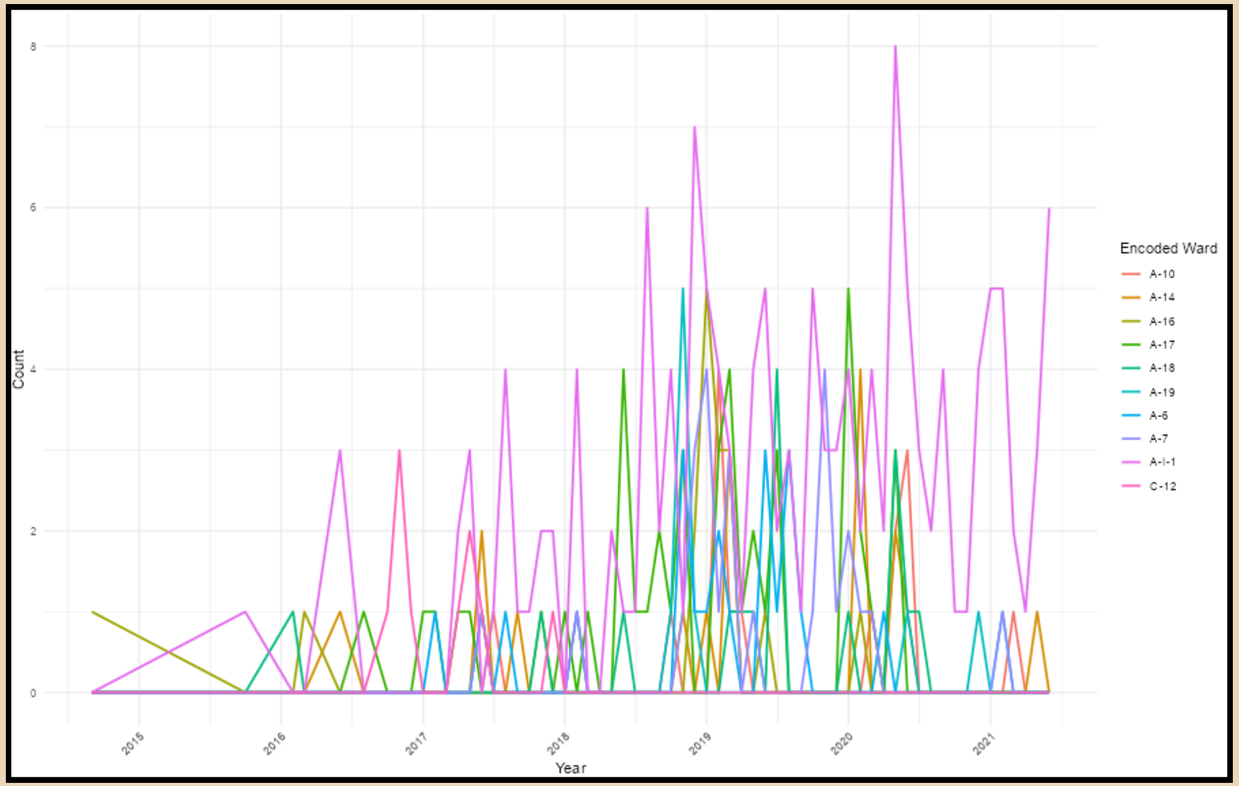
Observations:

- 226 observed ward-month combination (30 wards x 62 months)
- 88% of zero

Real Data

Characteristics:

- Total 30 wards
- Most of observations are close/equal to 0
- Highest: ICU (A-I-1)
- ICU (pink line): Dominate the others
- The number of cases is increasing overtime

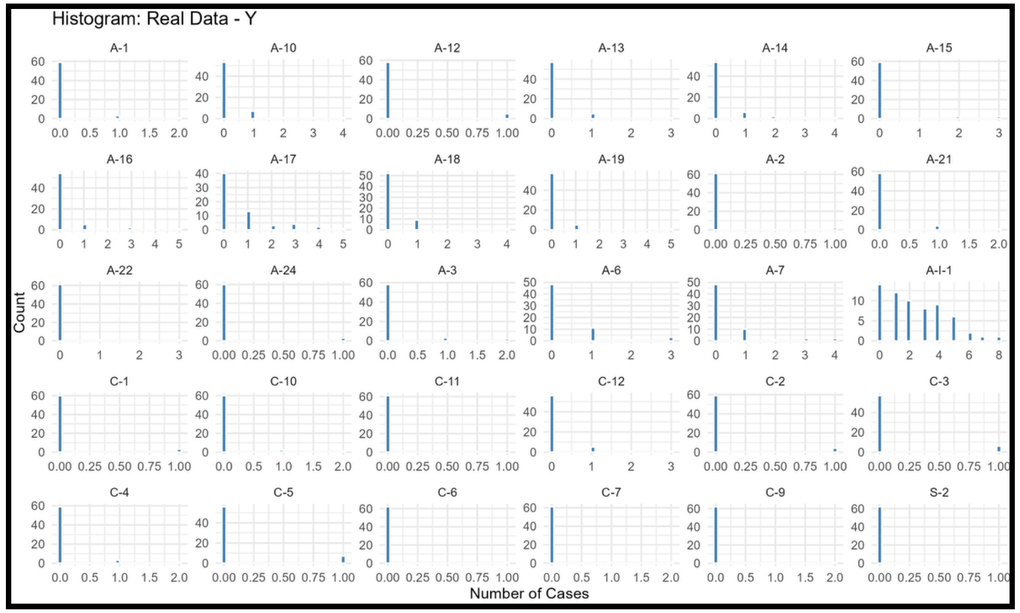


Line graph for top 10 wards with the highest total cases

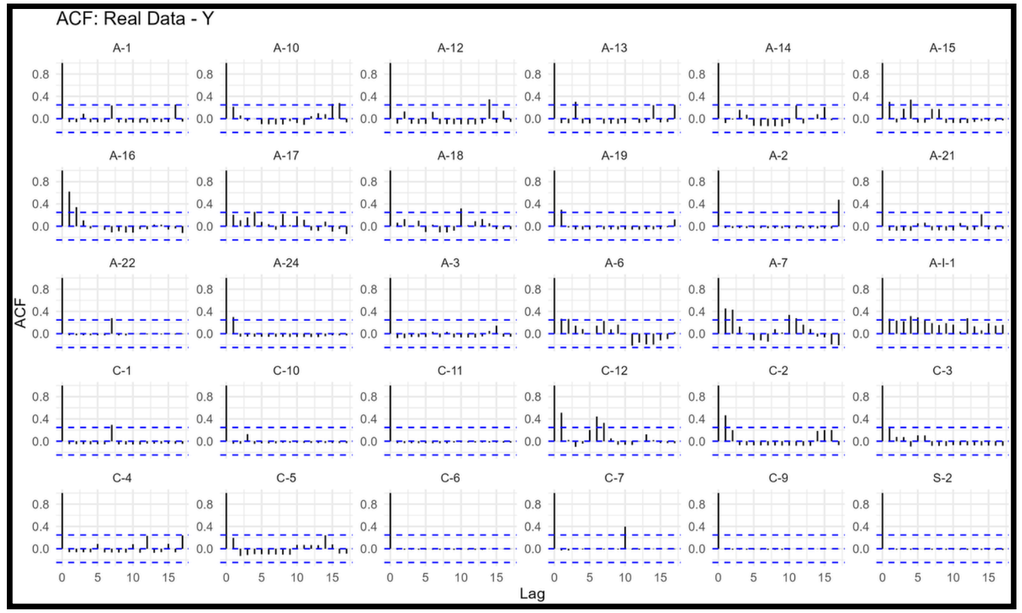
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Results

Real Data: (P =30, T = 100)



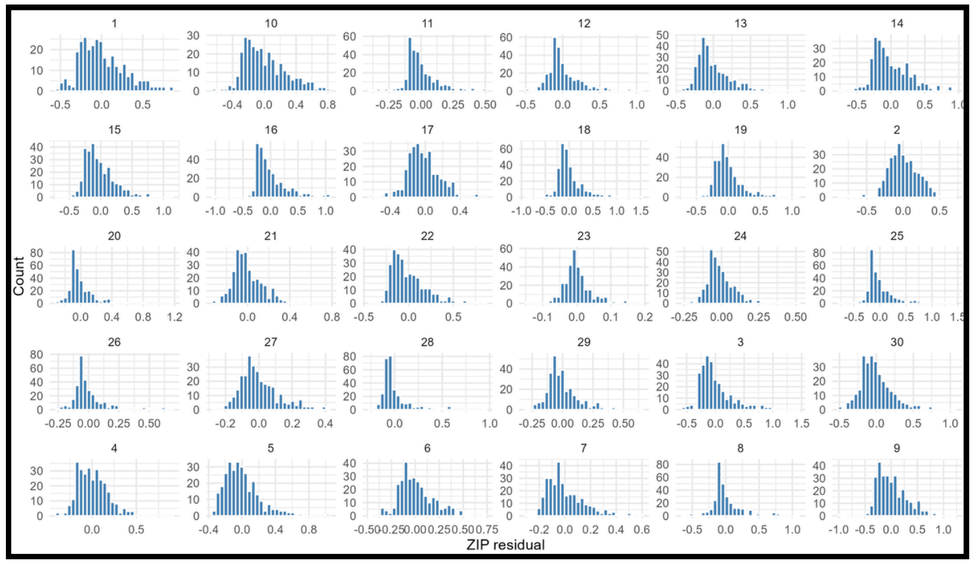
Real data: Histogram



Real data: ACF plot

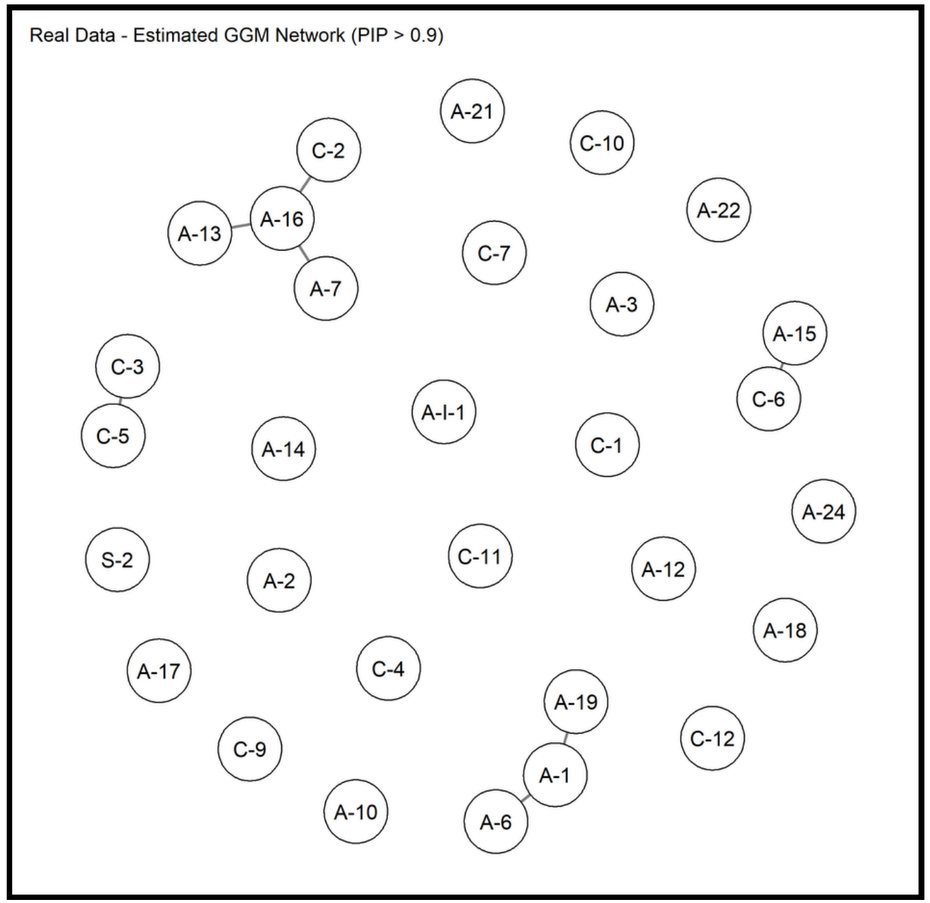
Results

Real Data: (P=30, T= 62)



Innovation latent: Histogram

GGM



Estimated graph (PIP > 90%)
(7 edges identified)

Results

Parameter	Posterior Mean	Posterior SD	95% CI
β_0	−0.995	0.707	[−2.234, 0.561]
β_0^* $(\beta_0 + c_S)$	−1.005	0.707	[−2.247, 0.554]
ρ	0.444	0.554	[−0.804, 0.997]
φ	0.462	0.535	[−0.761, 0.997]
c_Z	0.007	0.100	[−0.189, 0.203]
c_S	−0.010	0.100	[−0.206, 0.186]

Victorian CPE Surveillance Data Parameter Estimates

Baseline log-intensity

$$\widehat{\beta}_0^* = \widehat{\beta}_0 + \hat{c}_S = -1.005$$

- ≈ 0.366 cases per ward-month
- High spatial sparsity of CPE

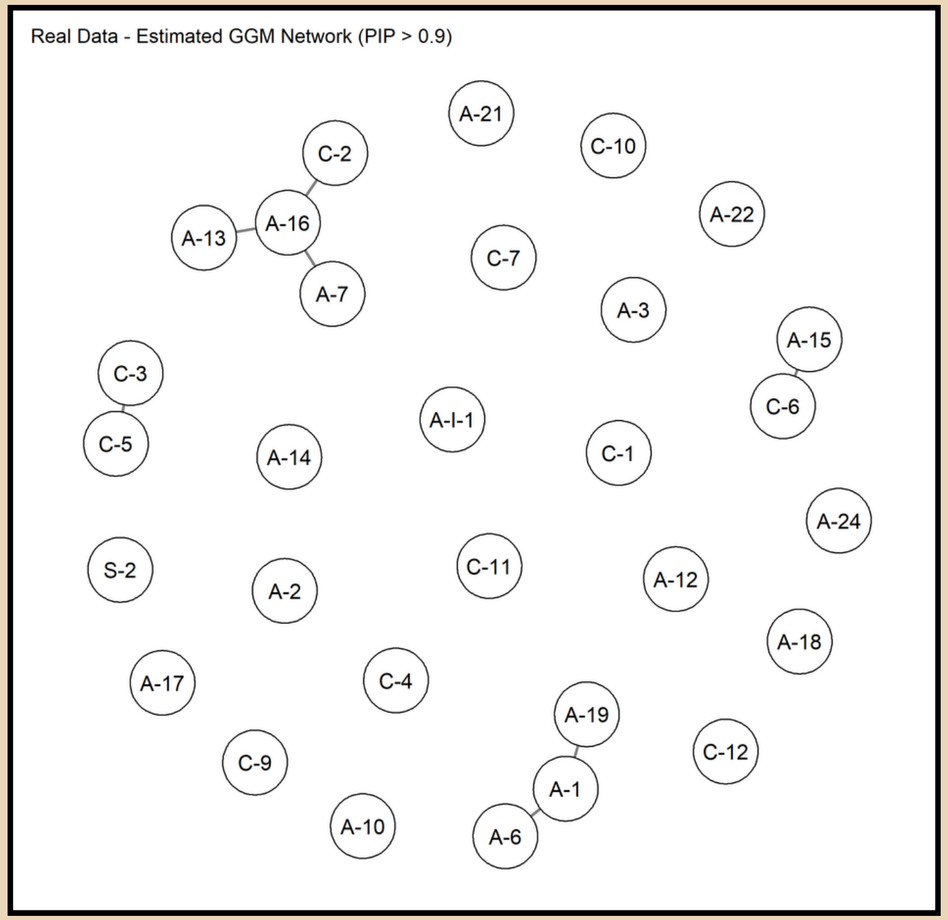
AR(1) coefficients

$$\widehat{\phi} = 0.462, \mathrm{SD} = 0.535$$
$$\widehat{\rho} = 0.444, \mathrm{SD} = 0.554$$

- Moderate positive temporal persistence

Interpretation

Estimated graph



Edges

- Only 7 edges over 435 possible edges
- Extreme sparsity aligns

→ Direct transmission pathways do not occur uniformly across all health facility wards, but are restricted to specific, highly interconnected clinical pathway (same first letter encoded)

Validation

- Simulation results (P=30 and T=100), 5 over 10 edges were recovered correctly (no false positive)

→ Be able to recover 50% of the true conditional dependence structure between wards

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Discussion

About this study

Extend Zero-Inflated Poisson Lognormal (ZIPLN) state-space to infer conditional dependence networks from sparse, zero-inflated, and **longitudinally correlated** counts.

Application

- Applied to Victorian CPE surveillance data across major health services (P=30, T=62).
- Provides actionable evidence for public health policy in Victoria (estimated graph)

Comparative Performance

Comparing with the BMGM model to demonstrate why accounting for temporal dynamics is essential (BMGM model)
→ Further direction to accommodate multivariate count data with mixed distributions (e.g. zero-inflated negative binomial)

Limitations and Future Directions

Current model:

- Low number of temporal observations relative to the number of wards (P=30, T=62)
- Prior specifications that are weakly informative

Future:

- Explore hierarchical or empirical Bayes approaches to learn prior hyperparameters from data

References

Batardère, Bastien, Julien Chiquet, François Gindraud, and Mahendra Mariadassou. 2025. “Zero-Inflation in the Multivariate Poisson Lognormal Family.” Statistics and Computing 35 (6): 196. https://doi.org/10.1007/s11222-025-10729-0.

Department of Health, Victoria. 2026. Victorian Guideline on CPO for Health Services. Https://www.health.vic.gov.au/infectious-diseases/victorian-guideline-on-cpo-for-health-services.

Florez, M., Gottard, A., McAdams, C., Guindani, M. and Vannucci, M. (2025), ‘A bayesian approach for inference on mixed graphical models’, Bayesian Analysis 1(1), 1–28.

Jewson, Jack, Li Li, Laura Battaglia, Stephen Hansen, David Rossell, and Piotr Zwiernik. 2024. “Graphical Model Inference with External Network Data.” Biometrics 80 (4): ujae151. https://doi.org/10.1093/biomtc/ujae151.

Wang, Hao. 2015. “Scaling It up: Stochastic Search Structure Learning in Graphical Models.” Bayesian Analysis 10 (2): 351–77. https://doi.org/10.1214/14-BA916.

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THANK YOU

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